



Introduction to Remote Sensing

Lab_08: Supervised image classification



402.467 / Applied GIS
Fundamentals

Agenda

➔ Introduction to satellites (basic characteristics)

➔ Introduction to Remote Sensing

➔ Earth Observation satellites, the example of Sentinel-2

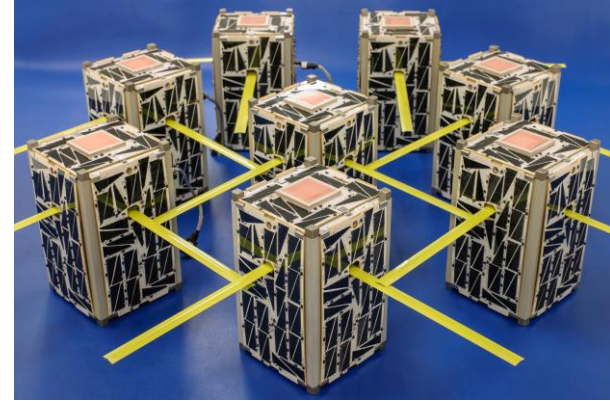
➔ Image classification

➔ Group projects



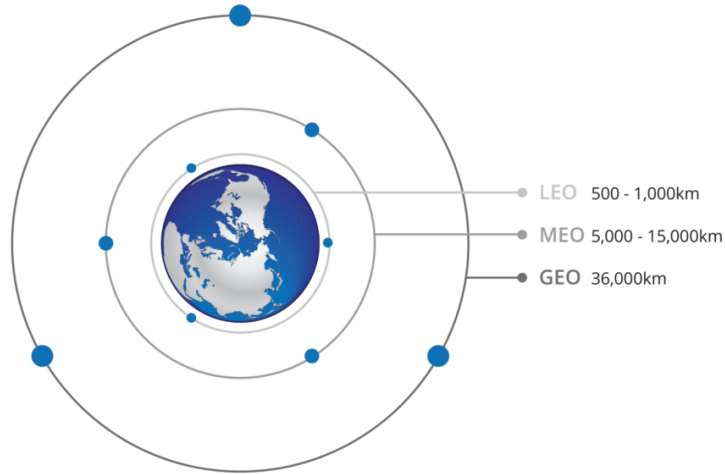
Introduction to satellites

CubeSats



- **Where do satellites orbit?**
- **What do satellites do?**
- **How many countries have satellites in orbit?**
- **What types of satellites are there?**
- **What is their size?**
- **How many EO satellites orbit the Earth as we speak?**

A satellite's orbit



Low Earth Orbit

Medium Earth Orbit

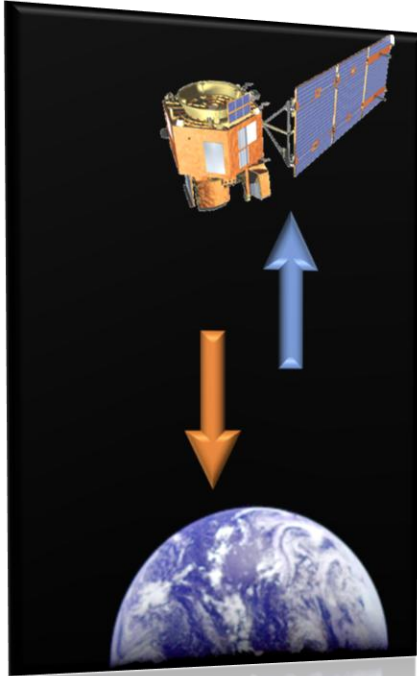
GEostationary Orbit



402.467 / Applied GIS
Fundamentals

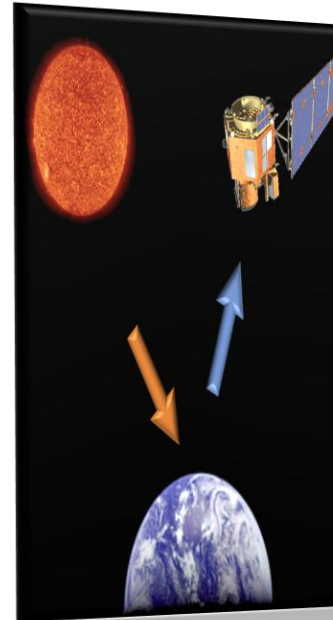
<https://nanoavionics.com/blog/how-many-satellites-are-in-space/>

Satellite types



Active sensors provide their own source of energy to scan an area and then detect and measure the radiation that is reflected back to the sensor. Examples are Synthetic Aperture Radar (SAR) systems and LiDAR (Light Detection and Ranging). Often used for terrain mapping and ice monitoring

Valdesera Christina Lydia
christina.valdesera@uni-graz.at



Passive sensors measure natural energy that is either reflected or emitted by the area being observed. Examples are optical cameras and multispectral scanners used on EO satellites like those in the Copernicus program

<https://www.nasa.gov/directorates/somd/space-communications-navigation-program/remotesensing/#:~:text=Remote%20sensing%20instruments%20are%20of,NASA>

10 December 2025

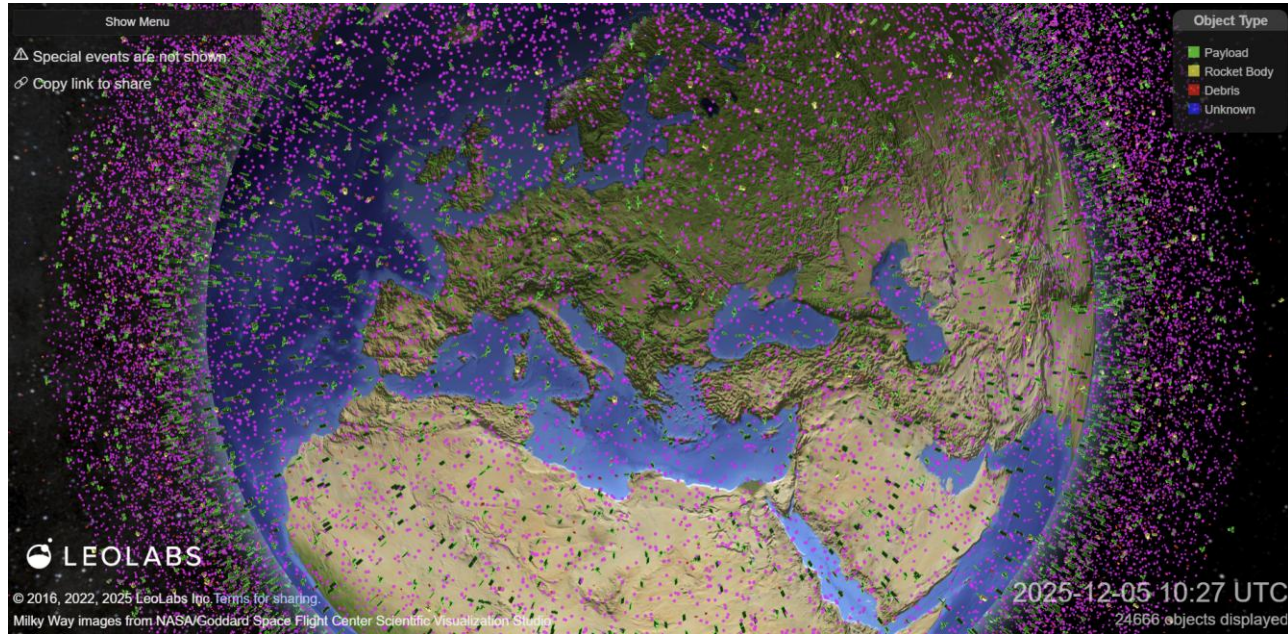
E0 satellites & applications

**Approximately 1,200
to over 1,500 active
EO satellites
&
at least 16 core
application areas**



Debris in space

There is more orbital debris than active satellites!



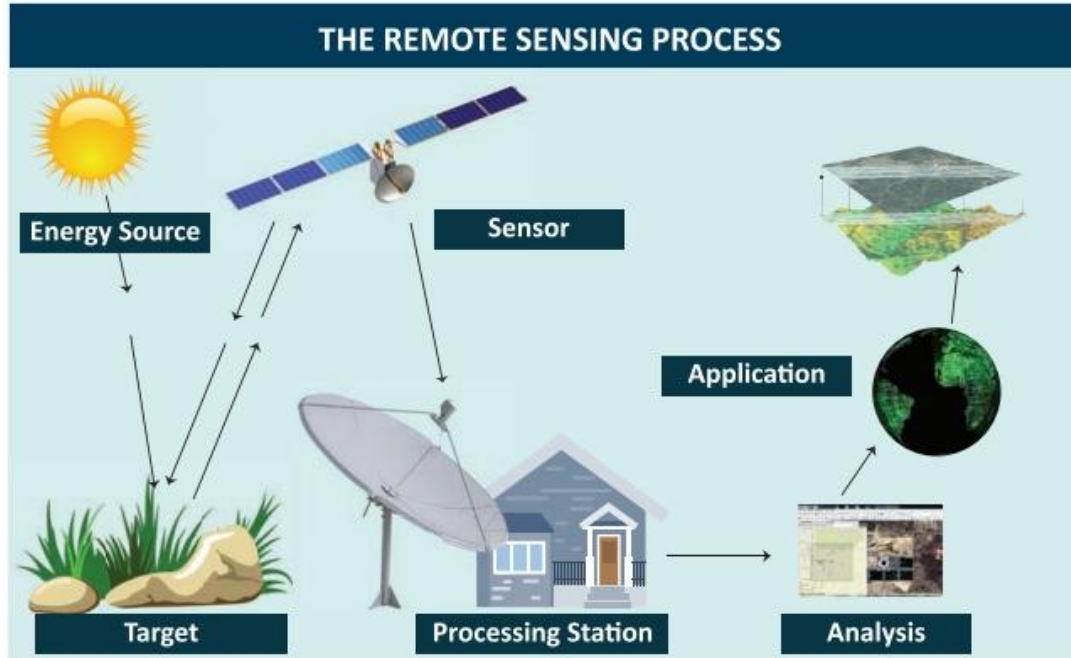
<https://platform.leolabs.space/visualizations/leo>

Satellites by mission & country

Mission	Number of Satellites
Communications	>8000
Earth Observation	>1000
Technology Development	>300
Navigation	>100
Space Science	>100

Country/Multinational Organization	Number of Operational Satellites
USA	>8000
Russia	>1500
China	>800
United Kingdom	>700
Japan	>200
France	>100
India	>100
Germany	>80
Canada	>70
Italy	>50

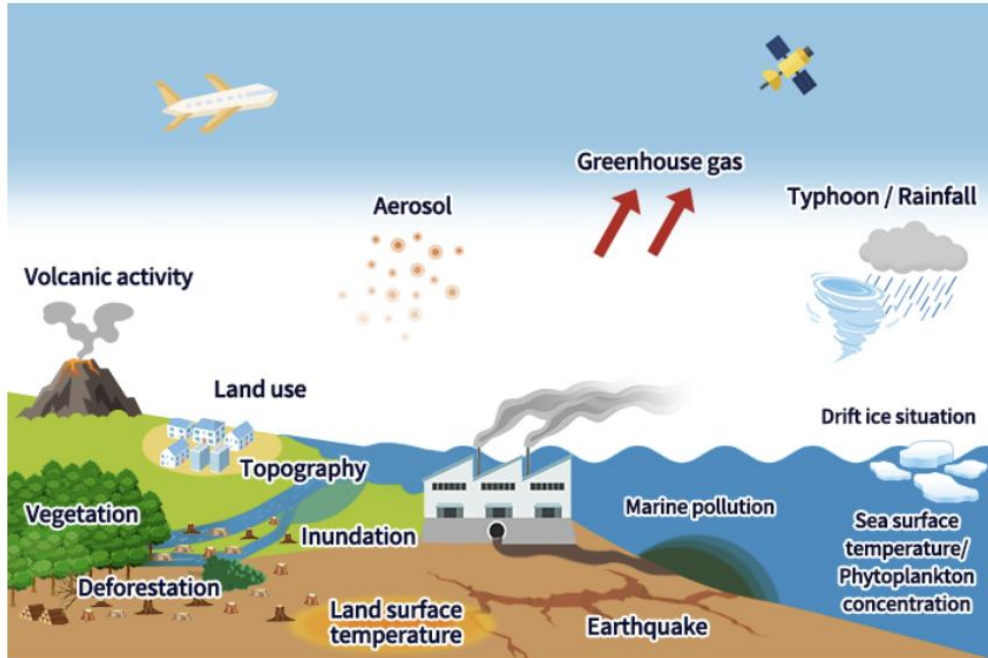
Introduction to Remote Sensing



- What can we find from EO satellite data?
- Where can we find EO data?
- The example of Sentinel-2 mission

Categories of EO by purpose

1. **Meteorology (weather forecasting)**
2. **Land use and agriculture**
3. **Oceanography**
4. **Climate & atmospheric monitoring**
5. **Disaster management**
6. **Terrain mapping & geology**



<https://earth.jaxa.jp/en/eo-knowledge/eosatellite-type/index.html#:~:text=Earth%20observation%20satellites%20are%20the,and%20color%20of%20oceans%2C%20etc>

Where can you get satellite images from

Free Government & Research Sources

- USGS EarthExplorer
- Copernicus Data Space Ecosystem / Sentinel Hub
- NASA Earthdata Search / Worldview
- Google Earth Pro



Commercial Providers

- Maxar Technologies
- Airbus Defence and Space
- Planet Labs
- UP42



Satellites & bands

Satellites like Sentinel 2 don't take normal color pictures; they use a special sensor called Multispectral Instrument or MSI that separates the light reflected from the Earth into several specific colors (called "bands"). This lets us identify different things on the ground, because everything from plants to water to buildings has a unique "light signature" or spectral response



Sentinel-2A (10m) Satellite Sensor



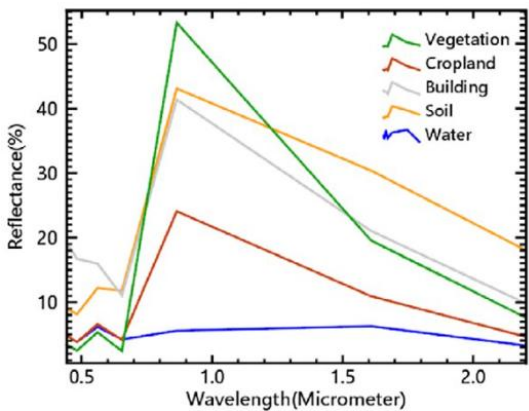
MSI

Sentinel-2 Bands	Central Wavelength (μm)	Resolution (m)
Band 1 - Coastal aerosol	0.443	60
Band 2 - Blue	0.490	10
Band 3 - Green	0.560	10
Band 4 - Red	0.665	10
Band 5 - Vegetation Red Edge	0.705	20
Band 6 - Vegetation Red Edge	0.740	20
Band 7 - Vegetation Red Edge	0.783	20
Band 8 - NIR	0.842	10
Band 8A - Vegetation Red Edge	0.865	20
Band 9 - Water vapour	0.945	60
Band 10 - SWIR - Cirrus	1.375	60
Band 11 - SWIR	1.610	20
Band 12 - SWIR	2.190	20

Image classification

How do satellites see and how do we interpret what they see?

1



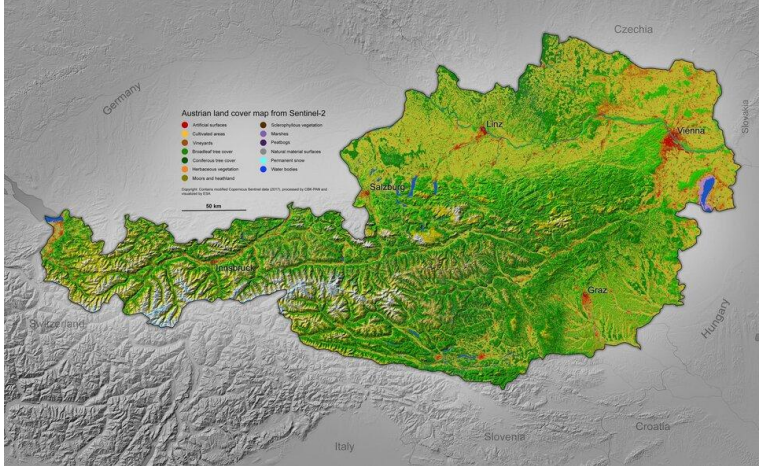
Spectral reflectance curves of vegetation, soil, water, etc.

2



Image classification

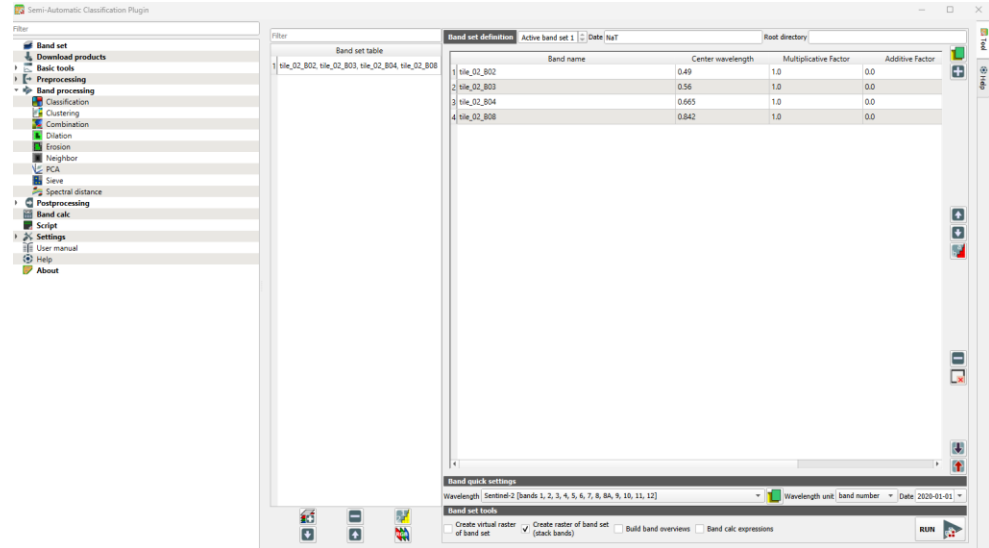
3



Land cover Austria

Semi-automatic classification plugin

- Pixel based image classification
- Supervised image classification
- Land cover image classification using SCP in QGIS



402-467 / Applied GIS
Fundamentals

https://semiautomaticclassificationmanual-v5.readthedocs.io/de/latest/tutorial_1.html?utm_source=chatgpt.com#tutorial-1-your-first-land-cover-classification

Group projects

 Indicative topic: «Quantify how much green space was lost to the expansion of the University Campus»

Guidelines

- ☐ Choose a topic that interests you
- ☐ Start small: neighbourhood to small city
- ☐ Look for open source data
- ☐ Document your methods (GIS report)
- ☐ Produce meaningful outputs (maps)

*You are totally free to choose the approach to conduct the analysis.
Feel free to reuse any of the data provided during the semester!
Everyone is expected to do some technical work!*

Evaluation details

1. GIS report (ca. 5-10 pages) 40%

- how effectively you explain the project's aim and the real-world impact
- data used & accuracy
- methods & tools used
- results & limitations
- references
- work distribution within the group & peer discussion

2. Presentation (10 mins + 5mins Q&A per team) 50%

- how effectively you communicate the above

3. Peer discussion 10%

- actively discuss the method and the results after the presentation
- offer thoughtful, constructive feedback on their methodologies and findings
- include it in the GIS report